Electron-Ion Path Integral Monte Carlo with the Pair-Product Action

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(Dated: May 6, 2026)

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Keywords: Path Integral; Monte Carlo; Quantum Mixture; Bosons; Fermions; Two Component Plasma; Thermodynamics; Structure; Superfluidity; Sign Problem; Metallic Hydrogen; Molecular Hydrogen; Hydrogen Atom; Hydrogen Molecule; Pair Coulomb Density Matrix

CONTENTS

I. Introduction 1

II. The computer experiment 1

A. Description of our PIMC and RPIMC algorithms 3

Author declarations 3
Conflicts of interest 3
Data availability 5
Funding 5

References 5

I. INTRODUCTION

Monte Carlo (MC) [1] electron-proton (ep) plasma as a fluid binary mixture [2]

II. THE COMPUTER EXPERIMENT

In order to study the quantum statistical properties of the ep fluid binary mixture (H) through a computer experiment it is necessary to add to our simulation box of volume Ω an equal number $N/2$ of electrons (e) of mass m_e and of protons (p) of mass m_p and let them interact with the usual Coulomb pair potential $\pm 1/r$. We thus obtain a fluid of number density $n = N/\Omega$. We can then study the properties of H in thermal equilibrium at an inverse temperature $\beta = 1/k_B T$ with k_B Boltzmann constant and T the absolute temperature. If the laws of quantum statistical physics and of Coulomb are meaningful we should be able to reproduce the thermodynamic and structural properties of H that we observe in nature.

In our reduced units we will measure lengths in units of a Bohr radius $a_B = \hbar^2/m_e e^2$ and energies in units of two Rydbergs e^2/a_B, i.e. of a Hartree. The Hamiltonian operator and the density matrix in reduced units will then read

$$H = K + V = -\sum_{\alpha=1}^{N} \lambda_{i_\alpha} \nabla_\alpha^2 + \sum_{\alpha<\beta=1}^{N} \phi_{i_\alpha i_\beta}(r_{\alpha\beta}), \tag{2.1}$$

$$\rho = e^{-H/T}. \tag{2.2}$$

where we indicate with a Greek index the particle label and with a Roman index the species label so that for example i_α is the species label of particle α, either 1 for the electrons or 2 for the protons, $\boldsymbol{r}_\alpha$ is the position vector of particle α, and $r_{\alpha\beta} = |\boldsymbol{r}_{\alpha\beta}| = |\boldsymbol{r}_\alpha - \boldsymbol{r}_\beta|$ is the distance between particles α and β. The partial pair potential is the usual Coulomb potential $\phi_{ij} = \epsilon_{ij}/r$ with $\epsilon_{ii} = +1$ for all i and $\epsilon_{ij} = -1$ for $i \neq j$. $\lambda_i = \hbar^2/2\mu_i$ where the reduced masses are $\mu_1 = 1$ and $\mu_2 = m_p/m_e \approx 1836.15$. Temperatures are in units of ≈ 315775K.

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The coordinate representation of the high temperature density matrix for distinguishable particles is [3]

$$\rho(R,R';\tau)=\rho_0(R,R';\tau)e^{-U(R,R';\tau)}, \tag{2.3}$$

$$\rho(R,R';\tau)=\prod_{\alpha=1}^{N}(4\pi\lambda_{i_\alpha}\tau)^{-3/2}e^{-(\boldsymbol{r}_\alpha-\boldsymbol{r}'_\alpha)^2/4\lambda_{i_\alpha}\tau}, \tag{2.4}$$

where $\tau=\beta/M$ for large M. So that the coordinate representation of the thermal density matrix for identical particles can be obtained from the following discretized path integral [4]

$$\rho(R,R';\beta)=\sum_{\mathscr{P}}\mathrm{sgn}(\mathscr{P})\int\rho(\mathscr{P}R,R_1;\tau)\rho(R_1,R_2;\tau)\cdots\rho(R_{M-1},R';\tau)\prod_{k=1}^{M-1}dR_k, \tag{2.5}$$

where $R_k=(\boldsymbol{r}_{1,k},\boldsymbol{r}_{2,k},\ldots,\boldsymbol{r}_{N,k})$ is the kth timeslice. Here $\boldsymbol{r}_{\alpha,k}$ is the position vector of particle α of species i_α at timeslice k in the path integral. Each high temperature density matrix gives a *link* between two successive *timeslices*. The sum is over the permutations $\mathscr{P}$ of identical particles (of the same species) and $\mathrm{sgn}(\mathscr{P})$ is the sign of the given permutation

$$\mathrm{sgn}(\mathscr{P})=\begin{cases}+1 & \text{Bose-Einstein statistics}\\ \begin{cases}+1 & \mathscr{P}\ \text{ an even permutation}\\ -1 & \mathscr{P}\ \text{ an odd permutation}\end{cases} & \text{Fermi-Dirac statistics}\end{cases} \tag{2.6}$$

While a Monte Carlo integration [1] of the path integral (2.5) (PIMC) can be carried out exactly for bosons, for fermions the alternating signs in the permutation sum becomes less and less efficient as N and/or M increase [5]. This is known as the 'fermions sign problem'. An approximate way to circumvent this problem is the Restricted Path Integral Monte Carlo (RPIMC) method of Ceperley [4, 5]. In our simulations we will always compare the PIMC for bosons and the RPIMC for fermions with a free fermions restriction.

For the inter-action U we will adopt the end-point approximation for the pair-product action [3]

$$U(R,R';\tau)\approx\sum_{\alpha<\beta}P_{i_\alpha i_\beta}(\boldsymbol{r}_{\alpha\beta},\boldsymbol{r}_{\alpha\beta};\tau), \tag{2.7}$$

where for P_{ij} we choose the pair Coulomb density matrix of Pollock [6]. Namely

$$P_{ij}(\boldsymbol{r},\boldsymbol{r}';\tau)=-\ln\left[\frac{\rho_{ij}(\boldsymbol{r},\boldsymbol{r}';\tau)}{\rho^0_{ij}(\boldsymbol{r},\boldsymbol{r}';\tau)}\right], \tag{2.8}$$

where

$$\frac{\partial\rho_{ij}(\boldsymbol{r},\boldsymbol{r}';\tau)}{\partial\tau}=\left[\lambda_{ij}\boldsymbol{\nabla}-\frac{\epsilon_{ij}}{r}\right]\rho_{ij}(\boldsymbol{r},\boldsymbol{r}';\tau). \tag{2.9}$$

Here $\lambda_{ij}=\lambda_i+\lambda_j$ and $\rho^0_{ij}(\boldsymbol{r},\boldsymbol{r}';\tau)=(4\pi\lambda_{ij}\tau)^{-3/2}\exp[-(\boldsymbol{r}-\boldsymbol{r}')^2/4\lambda_{ij}\tau]$ satisfies the same Eq. (2.9) but for $\epsilon_{ij}=0$ for all i,j.

In a PIMC computer experiment one needs to worry about two limiting procedures: i. the thermodynamic $N\to\infty$ limit and ii. the continuum $M\to\infty$ limit. The simulation box of volume $\Omega=L^3$ can be made periodic so that it permeates the whole infinite three dimensional space in order to mimic the thermodynamic limit. This will produce finite size errors that can be reduced by increasing the box size L. Of course in order to study a given thermodynamic state of the system this requires increasing L at constant density n which means increasing N. This will make the computer experiment more demanding from the numerical calculation cost point of view. Moreover, in periodic boundary conditions the potential energy V for the particles within the central box should include also the pair interactions with all their images in the periodic boxes. Since the Coulomb potential $1/r$ is long range these contributions cannot generally be neglected and specific techniques should be used [7] to treat them, the most common of which is the Ewald summation. In this work we will neglect interactions with periodic images and we use the end-point approximation for the pair-product action. The continuum limit requires an M sufficiently large for the characteristic size of a free link of each species be much smaller than the box side, i.e. $\sigma_i=\sqrt{2\lambda_i\tau}\ll L$ for all i, so that we can neglect the periodic copies of the kinetic action.

Note that one could introduce for each species some length scales

$$\mathscr{W}_i=n_i^{-1/3},\quad i=1,2 \tag{2.10}$$

and rescale each reduced coordinate as $\tilde{\boldsymbol{r}}_\alpha=\boldsymbol{r}_\alpha/\mathscr{W}_{i_\alpha}$. Then the rescaled kinetic energy of species i, $\tilde{K}_i=\mathscr{W}_i^2K_i$ and the rescaled potential energy of species i, $\tilde{V}_i=\mathscr{W}_iV_i$. We then see that as n_i increases V_i is penalized respect to K_i and that component tends to behave more like a "free" gas with less correlations. This is confirmed by our numerical experiments.

TABLE I. Cases treated in our simulations for the *ep* fluid mixture. We show the total number of particles N, the reduced number density $n = N/\Omega$, the reduced temperature T, the total kinetic energy per particle $e_K = \langle K \rangle / N$, the total potential energy per particle $e_V = \langle V \rangle / N$, the superfluid fraction for species i, $f_s(i)$. In the statistics column "b" stands for bosons, "f" for fermions. The simulation lasted more than 10^8 Monte Carlo steps.

case	statistics	T	M	n	N	Ne_K	$-Ne_V$	$f_s(1)$	$f_s(2)$
A	b (PIMC)	0.15	20	0.3	30	10.44(1)	20.88(1)	0.596(1)	0.04774(7)
B	f (RPIMC)	0.15	20	0.3	20	...	...	...	...
C	b (PIMC)	0.15	20	0.16	30	9.83(1)	19.55(2)	0.486(2)	0.04783(7)
D	f (RPIMC)	0.15	20	0.16	20	...	...	...	...

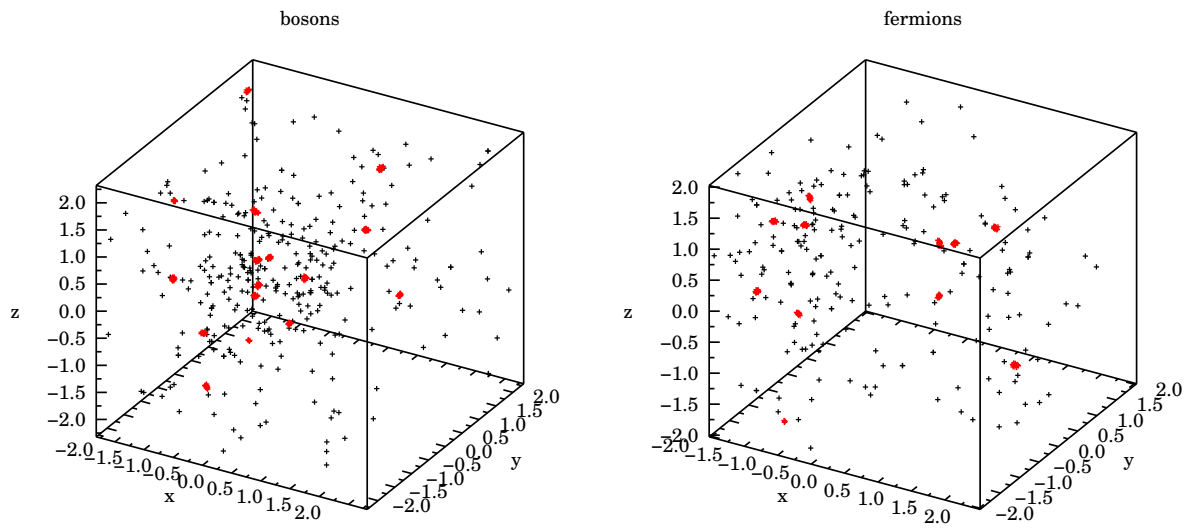

FIG. 1. Snapshots of the simulations for cases A (bosons) and B (fermions) at equilibrium. Red points are for the heavy slow protons and black points are for the light fast electrons. Each point represents the coordinates of the given species at each timeslice along the path. We see how bosons like to stay together while fermions like to stay apart.

Appendix A: Description of our PIMC and RPIMC algorithms

Given the order $N/2$ determinant

$$D_i^0(k_1, k_2) = \det \left|\left| (4\pi\lambda_i |k_1 - k_2| \tau)^{-3/2} e^{-(\boldsymbol{r}_{\alpha,k_1} - \boldsymbol{r}_{\beta,k_2})^2 / 4\lambda_i |k_1 - k_2| \tau} \right|\right|_{\alpha\beta}, \tag{A1}$$

such that $i_\alpha = i_\beta = i$, in the RPIMC with free particles restriction we reject the move whenever it produces a change of sign in the product $D_1^0(k_0, k) D_2^0(k_0, k)$ calculated between a reference timeslice k_0 (which may be chosen at random) and any other timeslice $k = 1, 2, \ldots, M-1$ [5]. Of course this method is exact only for free fermions. For our *ep* mixture it is just an approximation.

AUTHOR DECLARATIONS

Conflicts of interest

None declared.

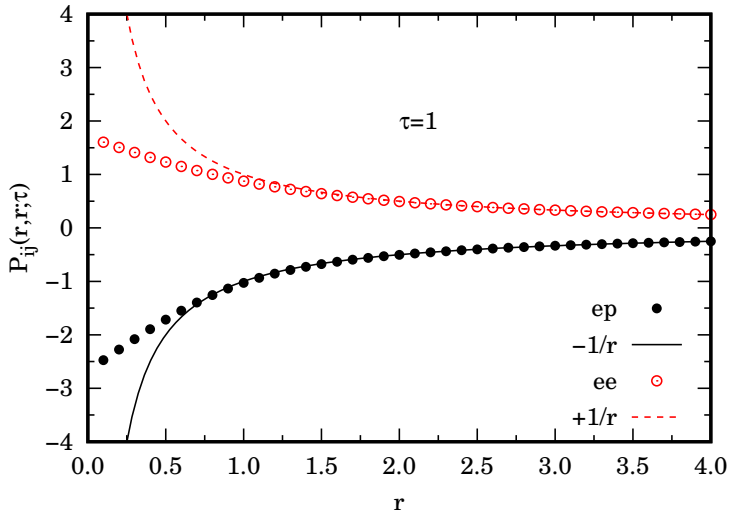

FIG. 2. We show the *ee* and *ep* pair Coulomb density matrix of Eq. (2.8) for $\tau = 1$. We compare it with the bare Coulomb potential. From the figure we can see that the cusp condition [6] $P_{ij}(\boldsymbol{r}, \boldsymbol{r}; \tau) \sim r\epsilon_{ij}/2\lambda_{ij}$ is satisfied at small r. The distance r is in units of a Bohr radius.

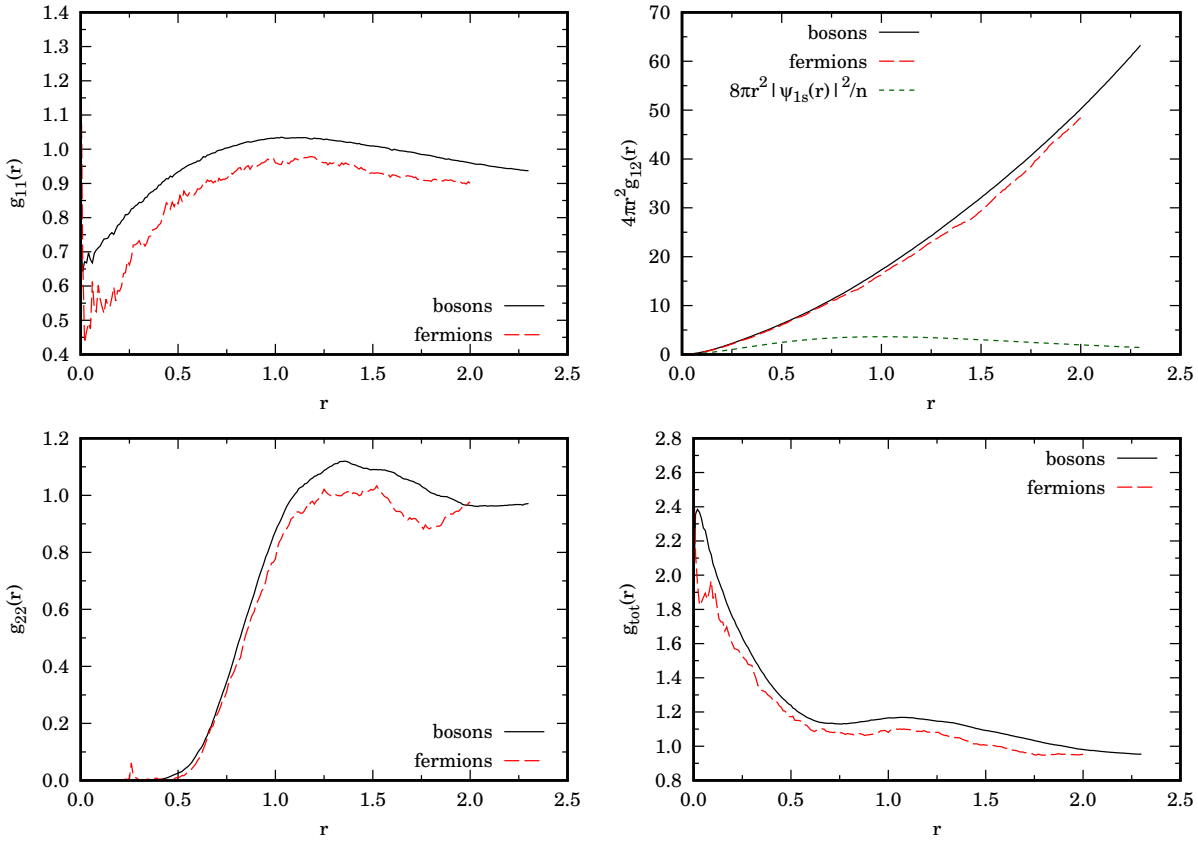

FIG. 3. We show the partial and total RDF of the H fluid for cases A and B in Table I. $\tau = 0.333$. In the panel for the unlike radial distribution function we also show the ground state probability distribution of the H atom $|\psi_{1s}|^2$ for comparison. The distance r is in units of a Bohr radius.

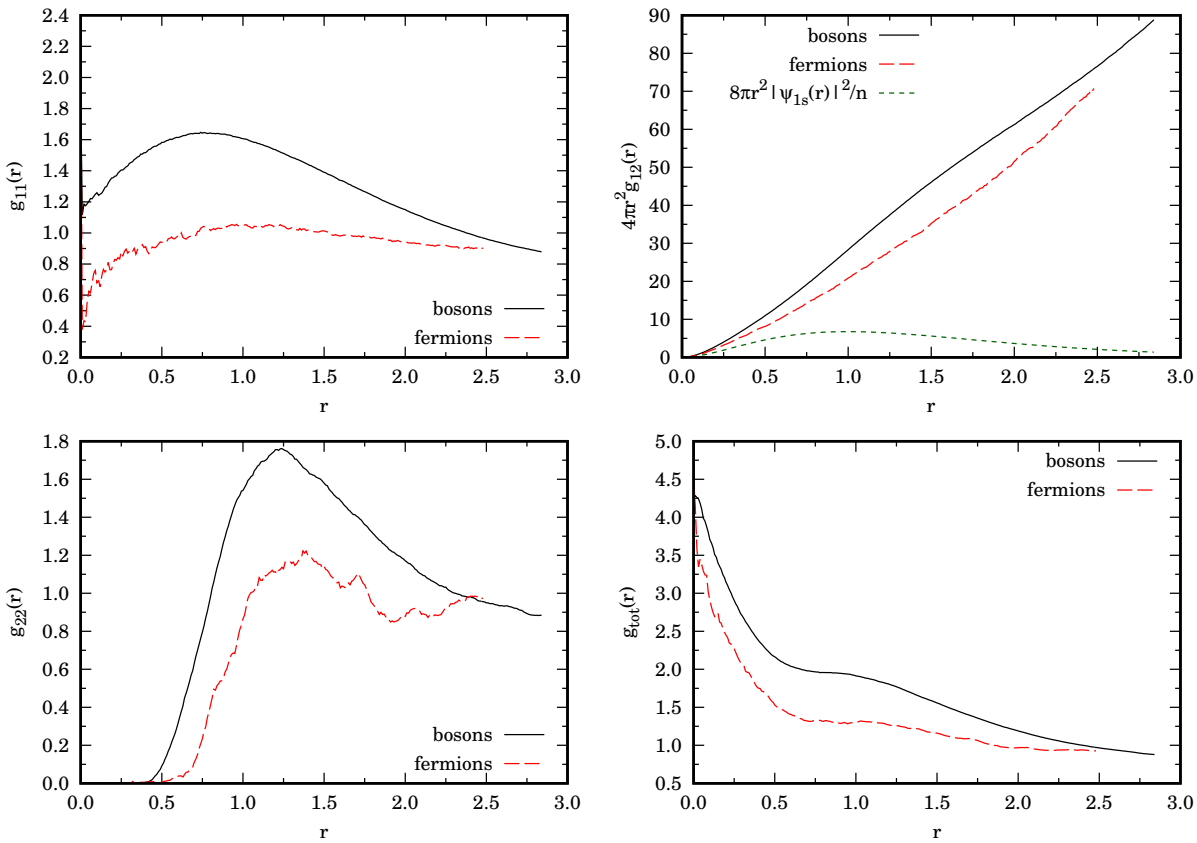

FIG. 4. We show the partial and total RDF of the H fluid for cases C and D in Table I. $\tau = 0.333$. In the panel for the unlike radial distribution function we also show the ground state probability distribution of the H atom $|\psi_{1s}|^2$ for comparison. The distance r is in units of a Bohr radius.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Funding

None declared.

[1] M. H. Kalos and P. A. Whitlock, *Monte Carlo Methods* (John Wiley & Sons Inc., New York, 1986).
[2] M. Bonitz et al., Towards first principles-based simulations of dense hydrogen, Physics of Plasma **31**, 110501 (2024).
[3] D. M. Ceperley, Path integrals in the theory of condensed Helium, Rev. Mod. Phys. **67**, 279 (1995).
[4] D. M. Ceperley, Path integral Monte Carlo methods for fermions, in *Monte Carlo and Molecular Dynamics of Condensed Matter Systems*, edited by K. Binder and G. Ciccotti (Editrice Compositori, Bologna, Italy, 1996).
[5] D. M. Ceperley, Fermion Nodes, J. Stat. Phys. **63**, 1237 (1991).
[6] E. L. Pollock, Properties and computation of the Coulomb pair density matrix, Comput. Phys. Commun. **52**, 49 (1988).
[7] V. Natoli and D. M. Ceperley, An Optimized Method for Treating Long-Range Potentials, Comput. Phys. **117**, 171 (1995).